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TITLE OF STUDY: Characterizing Quantitative Reasoning Across Contexts and Development
Principal Investigator: Michelle Hurst, PhD

The information in this consent form will provide more details about the research study. If you have any questions at any point, our research team is happy to answer them for you. After all your questions have been answered and you wish your child to take part in the research study, you will be asked to complete this permission form.

Who is conducting this study?

Michelle Hurst is the Principal Investigator of this research study and is conducting the study with her research team. A Principal Investigator has the overall responsibility for the conduct of the research. Michelle Hurst may be reached at michelle.hurst@rutgers.edu or 848-445-1601. This study is sponsored by the Eunice Kennedy Shriver National Institute of Child Health and Human Development, an institute of the National Institutes of Health.

Why is this study being done?

The goal of this study is to understand how people think about and describe different kinds of information, including how these change across early childhood.

Who may take part in this study and who may not?

Typically developing children within the target age range who are monolingual English or English dominant speakers may participate in this study.

How long will the study take?

The study session will last about X minutes [X depends on the version of the study the child is receiving and their age. This can range from 5-40 minutes, and will be filled in accordingly]. In most cases, we will only ask to see your child for one session, but sometimes we may ask your child to take part in additional sessions. If that is the case, we will contact you to ask about additional sessions and all future visits are optional.

What will my child and I be asked to do if they take part in this study?

During the session, your child will be asked to do some activities designed to assess knowledge and strategies for thinking about different kinds of information. These activities might involve comparing quantities, such as deciding which of two sets of dots is more or which of two game spinners with different probabilities is better for winning. Other activities might involve making decisions about pictures and shapes, such as matching shapes, deciding what direction a shape or picture is facing, assessing descriptions of pictures, or producing descriptions of scenarios.. Your child might also do activities involving age-appropriate language, math, and memory games, such as labeling pictures of objects, counting or arithmetic, and remembering a series of words or images. In addition, you might be asked to complete a survey about yourself, your child, and/or your family.

With your permission, we may video record your child's participation in this study. You do not have to provide permission to be recorded to take part in the main research. The recordings will be used for research and training purposes within our research lab. At the end of this form, we will also ask your permission to use the videos for research purposes outside the research team, educational purposes outside the research team, and for public purposes (e.g., including on our website). You do not have to provide permission for these optional uses. The recordings may include the following information that could identify you or your child: faces, names, and information about your home or your child's school in the background of the video (if applicable). [For school testing, this section will be removed if a school/off-site program will not allow recording]

What are the risks of harm or discomforts my child might experience by taking part in this study?

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We do not anticipate any harm or discomfort for your child by taking part in this study, beyond the risks encountered in daily life. There is a risk your child may be bored or frustrated by some of the activities. There is a risk of a breach of confidentiality of your child's participation or information, which we will minimize as described below.

Are there any benefits to my child if they take part in this study?

There are no known direct benefits from taking part in this study. However, many children find the studies fun and enjoy participating in them.

Will I receive the results of the research?

In general, we will not give you any individual results from the study. If you are interested in receiving a summary of the research results after completion of the study, please contact our research team at quadlab@psych.rutgers.edu.

Will my child be paid to take part in this study?

[Note: only one of the following will be used, depending on the testing location. Also, for short studies < 15 minutes or long studies > 45 minutes, we will adjust the payment accordingly, as described in our IRB protocol]

For children participating via Zoom: You will be given \$10 for participating in the study, in the form of an e-gift card to your email address. Your child will also receive a personalized certificate as a thank you.

For children participating via Lab: You will be given \$10 for participating in the study, in the form of an e-gift card to your email address or cash. Your child will also receive a personalized certificate and be able to select a small gift from our prize box as a thank you.

For children participating via schools/programs: Your child will receive a small gift (e.g., a sticker or small toy) as a thank you for participating. [NOTE: if the school does not allow us to provide these items, we will instead say: "Your child will not be provided compensation to take part in this study."]

How will information about my child be kept private or confidential?

Your child's personal information will be stored securely and separately from their responses in the study. Information that could identify your child will not be shared outside the research team and required oversight agencies (Rutgers Institutional Review Board, U.S. Department of Health and Human Services, and the National Institutes of Health) unless explicit permission is given (except as required by law, see Certificate of Confidentiality below). After information that could identify your child has been removed, de-identified information collected for this research may be made publicly available on data repositories (e.g., the Open Science Framework at <https://osf.io>) and used by us or others without obtaining additional permission from you. If you provide permission for video recording and your child's session is recorded, we will only share the video recordings in accordance with the privacy level you selected. All data (including recordings, if applicable) will be stored for a minimum of six years, and up to indefinitely, by the research team on secure cloud storage, password protected computers and hard drives, and in locked cabinets (in the case of physical or paper data).

Certificate of Confidentiality

This research is covered by a Certificate of Confidentiality from the National Institutes of Health. This means that the researchers cannot release or use information, documents, or samples that may identify you or your child in any action or suit unless you say it is okay. They also cannot provide them as evidence unless you have agreed. This protection includes federal, state, or local civil, criminal, administrative, legislative, or other proceedings. An example would be a court subpoena.

There are some important things that you need to know. The Certificate does not stop reporting that federal, state or local laws require. Some examples are laws that require reporting of child or elder abuse, some

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communicable diseases, and threats to harm yourself or others. The Certificate cannot be used to stop a sponsoring United States federal or state government agency from checking records or evaluating programs. The Certificate does not stop disclosures required by the federal Food and Drug Administration (FDA). The Certificate also does not prevent your information from being used for other research if allowed by federal regulations.

Researchers may release information about you or your child when you say it is okay. For example, you may give them permission to release information to insurers, medical providers or any other persons not connected with the research. The Certificate of Confidentiality does not stop you or your child from willingly releasing information about involvement in this research. It also does not prevent you or your child from having access to your/their own information.

What will happen if I do not wish my child to take part in the study or if I later decide that I do not wish my child to stay in the study?

It is your choice whether your child takes part in the research. You may choose to have your child take part, not to take part, or you may change your mind and withdraw your child from the study at any time. If you do not want your child to enter the study or decide to stop taking part, their relationship with the study staff will not change, and your child may do so without penalty. You may also withdraw your consent for the use of data already collected about your child, but you must do this in writing to Dr. Michelle Hurst (quadlab@psych.rutgers.edu or michelle.hurst@rutgers.edu). If the study data has already been de-identified and shared, you may not be able to withdraw your child's data.

Who can I call if I have questions?

If you have questions, concerns, problems, wish more information about your child's participation in the research, you can contact the Principal Investigator, Dr. Michelle Hurst, and her research team at quadlab@psych.rutgers.edu or 848-445-1601 [phone number might be changed to a lab line, once such a lab line exists].

If you have questions, concerns, problems, information or input about the research or would like to know your rights as a research subject, you can contact the Rutgers IRB or the Rutgers Human Subjects Protection Program via phone at (973) 972-3608 or (732) 235-2866 or (732) 235-9806 OR via email irboffice@research.rutgers.edu, or you can write us at 335 George Street, Liberty Plaza Suite 3200, New Brunswick, NJ 08901.

I have read this entire form, or it has been read to me, and I believe that I understand what has been discussed. All my questions about this form or this study have been answered. A copy of this form has been provided to me (or been made available to me to download).

Parent Name: [TYPE TEXT BOX for online forms]

Child's Name: [TYPE TEXT BOX for online forms]

Child's Date of Birth: [TYPE TEXT BOX for online forms, requiring date entry]

Today's Date: [TYPE TEXT BOX for online forms]

Parental Permission in Study Participation:

[] Yes, I agree to have me and my child participate in this study.

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☐ No, I do not agree to have me and my child participate in this study.

Parental Permission for Study Recording:

☐ Yes, I agree to have my child's session video recorded. I understand that this is not a requirement for participating in the study.

☐ No, I do not agree to have my child's session video recorded.

Privacy Levels for Study Recordings:

If you opted to allow for video recording, please select which (if any) optional uses of video you allow (please **select all that apply**):

☐ I give permission for the session recording or excerpts and images of the recording to be shown at scientific meetings or for educational purposes outside the research team. I understand that sometimes, these presentations are in public settings or may be recorded and that the recordings might become available online (e.g., YouTube). In giving my permission, I trust that authorized researchers will use their professional judgment and uphold ethical principles in determining which excerpts or images to present and to which audiences.

☐ I give permission to share data from this session with authorized researchers outside the research team in a secure library called Databrary. Authorized researchers may use the material for non-commercial research and educational purposes.

☐ I give permission for the research team to use the recordings, or excerpts/images from the recordings, in public facing ways, such as on their newsletter or website.